

BEYOND MEAN FOILS: AUDITING WORST-FOIL SPECIFICITY IN FROZEN CLIP REGION EXPLANATIONS

Kaixin Liu^{1,*}, Zhipeng Ye^{1,*†}, Feng Jiang¹, Qiufeng Wang², Zhenghao Wang³, Qihang Wu¹

¹Taizhou Institute of Science and Technology, Nanjing University of Science and Technology, Taizhou 225300, Jiangsu, China

²Department of Intelligence Science, Xi'an Jiaotong-Liverpool University, Suzhou 215123, Jiangsu, China

³Nanjing University of Science and Technology, Nanjing, China

24107880127@nustti.edu.cn, zhipengye@nustti.edu.cn, jf@nustti.edu.cn,

qiufeng.wang@xjtlu.edu.cn, wangzhenghao2002@outlook.com, 24107880128@nustti.edu.cn

*Equal contribution. †Corresponding author: Zhipeng Ye.

ABSTRACT

Spatial overlap and mean class contrast need not imply target-specific region responses. We audit CCI’s maximum-target-drop region in frozen CLIP, using separate prompts for selection and evaluation. Across COCO and VOC with two checkpoints, 41.08–64.78% of locality/mean-foil screen-passers fail a worst-foil criterion. With original screening memberships retained, excluding annotated-present foils leaves 39.69–63.64% failing. Exclusion reduces failure beyond exact cardinality-matched random removal by 0.86–0.92 percentage points on COCO; VOC intervals include zero. Positive-target filtering retains high failure. Among screened failures, unrestricted eight-region oracles permit sign repair in 6.25–7.84% on COCO and 27.40–31.15% on VOC. A target-drop tolerance of $\epsilon = .02$ reduces feasible repair to 0.16–0.98%; relaxing it increases opportunity with setting-dependent target and locality changes.

Index Terms— CLIP, explanation auditing, class specificity, constrained region selection

1 INTRODUCTION

CLIP [1] region explanations use intervention-induced image–text score changes. Cluster-based Concept Importance (CCI) [2] groups patches and measures their contribution; we audit its maximum-target-drop region, CCI-top1. A positive mean-foil contrast can average away a strong competitor: masking a region may reduce the target score more than the average non-target score, yet reduce one non-target score even more. Worst-foil specificity tests this strongest competitor, complementing spatial overlap and mean contrast.

Wang and Wang [3] analyze mean, max, and weighted class contrast. Their max contrast chooses the non-target class with the highest original-image logit; our worst foil maximizes the score drop under a region intervention and can vary with the region. CCI-top1 isolates the region selected for maximum target drop, enabling a direct audit of its cross-class specificity.

COCOA [4] explains corpus similarity relative to foil samples; CASE [5] tests class sensitivity and constructs contrastive saliency. Our contribution combines fixed-population auditing, exact cardinality-matched foil controls, and candidate/budget

decomposition to quantify worst-foil failure and decompose constraints on sign repair. Related CLIP attribution [6, 7, 8], class contrasts [9], and caption foils [10] provide complementary perspectives. WF reranking probes fixed CLIP candidates; IDEA [11] adapts CLIP for few-shot classification.

2 AUDIT AND REPAIR OPPORTUNITY

For checkpoint m , the pre-softmax score is $s_m(x, c) = \tau_m \langle z_m(x), e_m(c) \rangle$, with unit-norm embeddings and learned $\tau_m = \exp(\ell_m)$. Region R has intervention response

$$d_R(c) = s_m(x, c) - s_m(\mathcal{I}(x, R), c). \quad (1)$$

For target category t , selection uses “a photo of a {category}”:

$$R_{\text{CCI}} = \arg \max_{R \in \mathcal{R}(x)} d_R^{\text{sel}}(t). \quad (2)$$

Held-out margins and screens. Evaluation uses “a picture of the {category}”, “an image containing a {category}”, and “the {category}”. For unit-norm text embeddings $e_h(c)$, let $H = 3$, $n_c = \|H^{-1} \sum_h e_h(c)\|_2$, and $\bar{d}_c = H^{-1} \sum_h d_h(c)/n_c$. With $\mathcal{F}_t$ the non-target dataset categories, the aggregate worst-foil margin is

$$P(R, t) = \bar{d}_t - \max_{f \in \mathcal{F}_t} \bar{d}_f. \quad (3)$$

This equals using normalized mean text embeddings; class-specific n_c can change rankings. Tables report raw target drop $H^{-1} \sum_h d_h(t)$, whereas the positive-target screen uses $\bar{d}_t$. Selection and evaluation use separate templates.

For CCI-top1, define $A = \{\text{bbox} \geq .5, \bar{d}_t - \text{mean}_f \bar{d}_f > 0\}$ and $F = \{P < 0\}$. We use .5 as the locality threshold. To distinguish positive mean-foil contrast from positive target contribution, we also audit $A^+ = A \cap \{\bar{d}_t > 0\}$. The primary sign-repair population is $B = A \cap F$, fixed before reranking. Throughout, “failure” denotes $P < 0$, and “sign repair” denotes a transition from $P < 0$ to $P \geq 0$.

Target-constrained diagnostic probe. Define

$$M^{\text{sel}}(R, t) = d_R^{\text{sel}}(t) - \max_{f \in \mathcal{F}_t} d_R^{\text{sel}}(f), \quad (4)$$

$$\mathcal{R}_\epsilon = \{R : d_R^{\text{sel}}(t) \geq d_{R_{\text{CCI}}}^{\text{sel}}(t) - \epsilon\}, \quad (5)$$

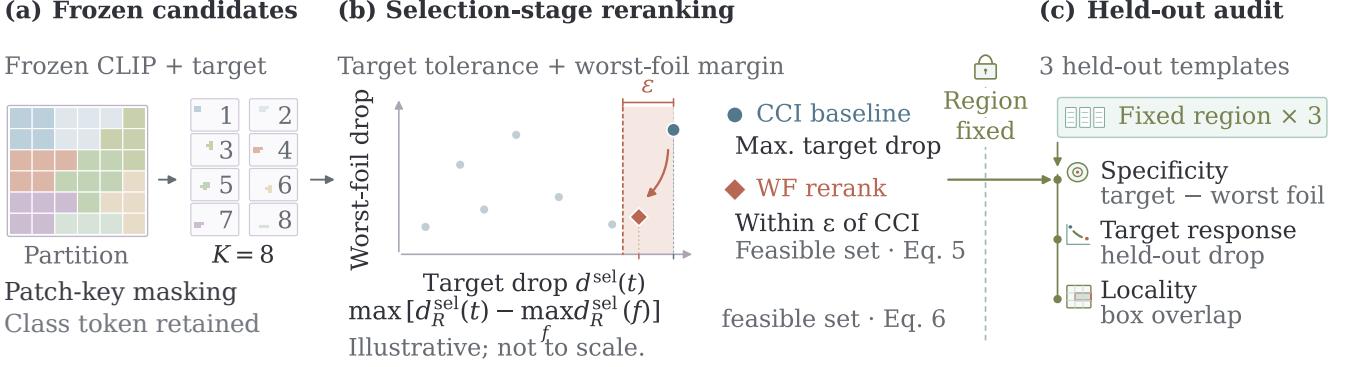


Figure 1: Selection and held-out audit. A region is fixed using selection-stage responses before three-template evaluation of specificity, target response, and locality. Schematic responses illustrate the selection rule.

and worst-foil (WF) reranking

$$R_{\text{WF}} = \arg \max_{R \in \mathcal{R}_\epsilon} M^{\text{sel}}(R, t). \quad (6)$$

The baseline is feasible, so any switch loses at most ϵ of *selection-stage* target drop. We measure its held-out effects separately. The reference budget is $\epsilon = .02$ logit units. A post-hoc .01/.02/.05/.10/.20 sweep measures budget sensitivity within each checkpoint; unrestricted oracles search the full candidate set.

Exact diagnostic oracles. For the screened-failure population B , the feasible sign-repair count is

$$N_O(B, \epsilon) = \sum_{i \in B} \mathbf{1} \left\{ \max_{R \in \mathcal{R}_\epsilon(i)} P_i(R) \geq 0 \right\}. \quad (7)$$

The unrestricted oracle searches all K candidates. A sign pass means $P_i(R) \geq 0$, without locality or positive-target constraints. Failures split into: no passing candidate (C_0); passing candidates but *none feasible* (C_1); a feasible pass missed by WF (C_2); and WF sign repair (r). Thus $|B| = C_0 + C_1 + C_2 + r$ and $N_O = C_2 + r$. We additionally count repairs satisfying $J(R) = \{P_i(R) \geq 0, A(R), \bar{d}_t(R) > 0\}$; $A(R)$ applies the same full-foil screen to the new region, without changing fixed baseline B . Oracles quantify held-out opportunity within the candidate set.

Fixed-population foil controls. For image i , let $\mathcal{F}_i^{\text{abs}}$ remove all annotated-present categories from the non-target foils. The protocol holds R_{CCI} and full-foil screen A fixed; F_i^{abs} tests $\bar{d}_t - \max_{f \in \mathcal{F}_i^{\text{abs}}} \bar{d}_f < 0$. Reducing the foil set weakly decreases failure, so we also match its cardinality. For N_i full foils, $k_i = |\mathcal{F}_i^{\text{abs}}|$, and m_i foils exceeding the target, a uniform size- k_i subset without replacement fails with probability

$$q_i = 1 - \frac{\binom{N_i - m_i}{k_i}}{\binom{N_i}{k_i}}. \quad (8)$$

With zero numerator when $k_i > N_i - m_i$, we compare $\Delta_A = |A|^{-1} \sum_{i \in A} (\mathbf{1}\{F_i^{\text{abs}}\} - q_i)$. Restricted sets are nonempty, and crowd annotations are excluded.

3 PROTOCOL AND REPRODUCIBILITY

COCO val2014 [12] and VOC2007 test [13] provide 27,708/1,943 eligible images, each evaluated with OpenAI CLIP B/16 and B/32 (59,302 image–model records). The target is the largest instance’s category, using annotation area on COCO and box area on VOC. We require another category and 2–80% target area; COCO crowd annotations are excluded. Full foils are the other 79/19 classes.

K-means clusters L2-normalized projected patch features: $K = 8$, three initializations, 50 iterations, seed 1701 + i (sorted IDs shuffled with seed 1701 define i). Intervention $\mathcal{I}$ adds $-\infty$ to attention logits at selected patch-key columns before softmax in every block/head, retaining the class-token column. Ties use first argmax; MPS features are cast to float32. BBox precision is the fraction of selected patch centers in target-class box unions, mapped by $224/\min(W, H)$ and center-crop offsets [14].

The primary results use a single traceable release: candidate masks, selection responses, three-template held-out responses, and bbox records were saved in the same generation pass. The validation checks patch partitions, bbox recomputation, response aggregation, candidate choices, and the failure decomposition. All primary tables use this release; Table 4 separately retains the earlier-run audit population.

Intervals use 10,000 image bootstraps and percentile 2.5/97.5 limits, pointwise without multiplicity correction. Paired quantities are differenced within image before re-sampling. Foil comparisons resample within fixed A/A^+ ; repair-cost intervals condition on actual repairs. Exact q_i is primary; 100 seeded draws check it.

4 RESULTS

Table 1: Fixed baseline A : full-foil, annotation-absent, and exact cardinality-matched random failure rates (%). Δ_A is absent minus random in percentage points, with paired 95% CI.

Setting	Full	Absent	Random	Δ_A
COCO B/16	64.37	63.17	64.09	-0.92 [-1.08,-0.77]
COCO B/32	64.78	63.64	64.50	-0.86 [-1.01,-0.71]
VOC B/16	42.27	40.94	41.46	-0.52 [-1.16,+0.06]
VOC B/32	41.08	39.69	40.24	-0.55 [-1.21,+0.04]

Failure persists after annotation exclusion. The fixed screen A contains 17,726, 17,848, 1,209, and 1,227 images in table order. Exclusion lowers failure by only 1.14–1.39 percentage points of A ; 96.63–98.24% of fixed baseline failures B remain (Table 1). Table 1 separates this change from foil-count reduction: COCO exclusion has a larger effect than random removal of the same number of categories, whereas VOC Δ_A intervals include zero. The residual failure extends beyond the contribution of explicitly annotated co-occurring classes.

Positive-target and scene-count checks. The mismatch persists under positive-target filtering. Among A^+ , full-foil failure is 62.64–63.79% on COCO and 40.66–41.40% on VOC. After annotation exclusion, it remains 61.38–62.62% and 39.26–40.05%, respectively. Matched-random comparisons retain the COCO/VOC interval pattern. Even scenes with exactly two annotated categories have COCO annotation-absent failure of 63.39/64.95%.

Table 2: Exact local decomposition of *screened failures* $B = A \cap F$ at $\epsilon = .02$. Counts in C_0, C_1, C_2, r are mutually exclusive; $C_2 + r$ is the feasible-oracle count.

Setting	$ B $	C_0	C_1	C_2	r
COCO B/16	11410	10697	695	0	18
COCO B/32	11562	10655	876	1	30
VOC B/16	511	371	135	0	5
VOC B/32	504	347	154	1	2

Candidate and budget bottlenecks differ by dataset. No candidate passes for 92.16–93.75% of COCO screened failures B , versus 68.85–72.60% on VOC (Table 2). Unrestricted sign-repair capacity is thus 6.25–7.84% of COCO B and 27.40–31.15% of VOC B (Figure 2). The .02 budget excludes 96.43–98.09% of these opportunities, leaving feasible capacity of just 0.16–0.98% of B . WF misses at most one feasible opportunity per setting (C_2 in Table 2).

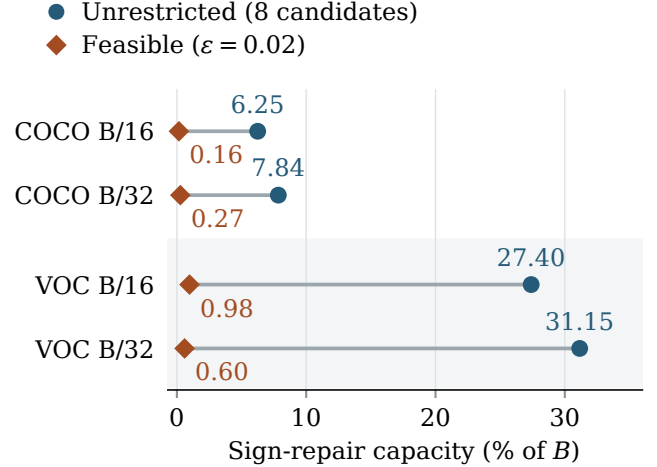


Figure 2: Budget-constrained sign-repair capacity among fixed screened failures B . Connected markers compare the same setting on a shared linear scale. Unrestricted: at least one of all eight candidates has $P \geq 0$; feasible: at least one also meets $\epsilon = .02$. Percentages use each setting’s own $|B|$ (Table 2) and are computed by exact enumeration.

Table 3: Budget endpoints (C: COCO; V: VOC). Multi: percentage of all eligible images with multiple feasible candidates. O_B, r_B : oracle/actual sign-repair percentages of screened failures B . $\Delta d_t, \Delta b$: all-image raw-target and bbox changes (bbox in percentage points).

Set	ϵ	Multi	O_B	r_B	Δd_t	Δb
C16	0.02	4.71	0.16	0.16	+0.0001	-0.12
C16	0.20	35.78	1.57	1.50	-0.0140	-1.63
C32	0.02	4.40	0.27	0.26	-0.0001	-0.08
C32	0.20	34.95	2.05	2.00	-0.0143	-0.60
V16	0.02	4.79	0.98	0.98	-0.0008	-0.34
V16	0.20	34.48	6.65	6.07	-0.0223	+0.08
V32	0.02	4.73	0.60	0.40	-0.0007	+0.02
V32	0.20	34.64	7.14	6.75	-0.0161	+1.07

Budget relaxation helps in every setting. At .02, 95.21–95.60% of images have only one feasible candidate (Table 3); WF has little room to change the region. At .20, oracle sign-repair capacity rises to 1.57–2.05% of COCO B and 6.65–7.14% of VOC B . Raw-target changes become negative in all four settings, but bbox changes are negative on COCO and positive on VOC. Budget relaxation therefore has dataset-dependent locality effects. Each comparison uses the same checkpoint and fixed baseline population.

Population-wide and sign-repair effects. Table 3 includes unchanged regions. In contrast, Table 5 conditions on actual sign repairs within fixed baseline B . The two tables describe population-wide changes and changes among successful sign repairs, respectively.

Table 4: Normalization sensitivity on fixed earlier-run A_{old} and earlier-run CCI regions. Raw/Norm: failure rates (%); Flip: percentage whose failure status differs. This audit uses the earlier-run population.

Setting	$ A_{\text{old}} $	Raw	Norm	Flip
COCO B/16	17,726	64.318	64.369	0.107
COCO B/32	17,847	64.762	64.778	0.073
VOC B/16	1,209	42.184	42.266	0.083
VOC B/32	1,227	41.157	41.157	0.000

Normalization has a small effect on the fixed audit. With earlier-run regions and A_{old} held fixed, raw and normalized failure rates differ by at most 0.083 percentage points; sign flips affect at most 0.107% of this population (Table 4). The hardest foil agrees on 99.42–99.84% of images. These paired comparisons preserve the high failure rate under both score definitions.

Table 5: Actual sign repairs *within fixed baseline B*, $\epsilon = .02$. r counts repairs; j counts those also satisfying J . WF–CCI means with repair-subset 95% CIs. Δd_t is raw held-out target-score change; only bbox change is in percentage points.

Setting	r	j	Δd_t	100 Δbbox
COCO B/16	18	12	0.029	-19.53
			[-0.096,0.148]	[-38.12,-1.32]
COCO B/32	30	25	0.094	-14.04
			[0.036,0.148]	[-27.70,-1.74]
VOC B/16	5	3	-0.021	-16.87
			[-0.170,0.182]	[-26.11,-7.00]
VOC B/32	2	1	-0.078	-28.17
			[-0.115,-0.042]	[-78.57,22.22]

Target and locality changes after sign repair. In every setting, some sign repairs fail to retain the locality/mean-foil screen and positive target contribution ($j < r$ in Table 5). The joint feasible-oracle counts are 12/26/3/2. VOC estimates are based on small sign-repair subsets. Outside the screened population, sign transitions can also regress: COCO B/32 has four opposite-sign regressions among all baseline passes. Counts over all failures or all switched regions have different denominators from the screened sign repairs reported here.

Table 6: Nonzero-threshold sensitivity on fixed baseline A . Entries are percentages of screen-passers A with $P < -\delta$ using the normalized aggregate margin. Thresholds are evaluated post hoc in logit units.

Setting	$\delta = 0$	.01	.10	.50
COCO B/16	64.37	63.94	59.49	36.05
COCO B/32	64.78	64.38	60.22	35.19
VOC B/16	42.27	41.44	36.23	13.73
VOC B/32	41.08	40.26	34.39	11.90

Failure persists at larger margin thresholds. At $\delta = .10$, 59.49–60.22% of COCO and 34.39–36.23% of VOC screen-passers A still fail (Table 6). Rates fall further at .50, so the magnitude and threshold matter. The threshold sweep characterizes the magnitude of failure within the normalized aggregate score.

5 INTERPRETATION AND LIMITATIONS

Interpretation. The audit identifies a persistent gap between locality/mean-foil screening and worst-foil specificity. Matched-cardinality controls separate annotation exclusion from foil-count reduction. Candidate enumeration attributes limited sign-repair capacity to candidate availability and the target-drop budget, with few additional opportunities missed by WF.

Scope. The results concern CCI-top1, eight candidates, and the specified attention intervention. COCO/VOC differences combine foil count, target selection, and annotation conditions. Other candidate constructions and area-matched controls require further experiments. Bbox precision measures patch-center overlap. Semantic interpretation of hard foils requires annotation review and human validation. Small VOC sign-repair subsets limit precision of conditional endpoint estimates.

Traceable evidence. The frozen release provides candidate summaries, validation reports, configuration, and hashes. Same-generation masks and full responses are retained locally. Three images changed selected regions from the earlier run. Primary results use the new records; Table 4 fixes the earlier-run regions and population. Prompt holdout covers templates within the same images and models; cross-checkpoint budget calibration requires further evaluation.

6 CONCLUSION

Locality and positive mean-foil contrast leave substantial worst-foil failure in CCI-top1 regions, persisting after annotation exclusion and at larger margin thresholds. Candidate availability and target-drop constraints account for most limits on sign repair. Larger budgets increase capacity with setting-dependent target and locality effects. These findings support reporting worst-foil specificity alongside locality and mean-foil contrast when evaluating frozen CLIP region explanations.

REFERENCES

- [1] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, “Learning transferable visual models from natural language supervision,” in *Proceedings of the 38th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 139. PMLR, 2021, pp. 8748–8763. [Online]. Available: <https://proceedings.mlr.press/v139/radford21a.html>.
- [2] A. Agarwal, S. Karanam, and V. Gandhi, “Concept regions matter: Benchmarking CLIP with a new cluster-importance approach,” *arXiv preprint arXiv:2511.12978*, 2025. [Online]. Available: <https://arxiv.org/abs/2511.12978>.
- [3] Y. Wang and X. Wang, “‘Why not other classes?’: Towards class-contrastive back-propagation explanations,” in *Advances in Neural Information Processing Systems*, vol. 35, 2022. [Online]. Available: <https://doi.org/10.52202/068431-0660>.
- [4] C. Lin, H. Chen, C. Kim, and S.-I. Lee, “Contrastive corpus attribution for explaining representations,” in *International Conference on Learning Representations (ICLR)*, 2023. [Online]. Available: <https://arxiv.org/abs/2210.00107>.
- [5] D. Williamson, Y. Ji, and M. Dwyer, “CASE: Contrastive activation for saliency estimation,” *arXiv preprint arXiv:2506.07327*, 2025. [Online]. Available: <https://arxiv.org/abs/2506.07327>.
- [6] Y. Wang, T. G. J. Rudner, and A. G. Wilson, “Visual explanations of image-text representations via multi-modal information bottleneck attribution,” in *Advances in Neural Information Processing Systems*, vol. 36, 2023, pp. 16 009–16 027. [Online]. Available: <https://doi.org/10.52202/075280-0704>.
- [7] Y. Li, H. Wang, Y. Duan, J. Zhang, and X. Li, “A closer look at the explainability of contrastive language-image pre-training,” *Pattern Recognition*, vol. 162, p. 111409, 2025. [Online]. Available: <https://doi.org/10.1016/j.patcog.2025.111409>.
- [8] M. Dreyer, L. Hufe, J. Berend, T. Wiegand, S. Lapuschkin, and W. Samek, “From what to how: Attributing CLIP’s latent components reveals unexpected semantic reliance,” *arXiv preprint arXiv:2505.20229*, 2025. [Online]. Available: <https://arxiv.org/abs/2505.20229>.
- [9] F. Li, Z. Meng, C. Tan, and Y. Yin, “Class-disentangled attribution with gradient guidance for CLIP,” *Journal of King Saud University Computer and Information Sciences*, vol. 38, p. 376, 2026. [Online]. Available: <https://doi.org/10.1007/s44443-026-00779-3>.
- [10] M. Yuksekgonul, F. Bianchi, P. Kalluri, D. Jurafsky, and J. Zou, “When and why vision-language models behave like bags-of-words, and what to do about it?” in *International Conference on Learning Representations (ICLR)*, 2023. [Online]. Available: <https://openreview.net/forum?id=KRLUvvh8uaX>.
- [11] Z. Ye, F. Jiang, Q. Wang, K. Huang, and J. Huang, “IDEA: Image description enhanced CLIP-adaptor for image classification,” *Pattern Recognition*, vol. 171, p. 112224, 2026. [Online]. Available: <https://doi.org/10.1016/j.patcog.2025.112224>.
- [12] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollar, and C. L. Zitnick, “Microsoft COCO: Common objects in context,” in *Computer Vision—ECCV 2014*, ser. Lecture Notes in Computer Science, vol. 8693. Springer, 2014, pp. 740–755. [Online]. Available: https://doi.org/10.1007/978-3-319-10602-1_48.
- [13] M. Everingham, L. Van Gool, C. K. I. Williams, J. Winn, and A. Zisserman, “The PASCAL visual object classes (VOC) challenge,” *International Journal of Computer Vision*, vol. 88, no. 2, pp. 303–338, 2010. [Online]. Available: <https://doi.org/10.1007/s11263-009-0275-4>.
- [14] T. Gomez, T. Freour, and H. Mouchere, “Metrics for saliency map evaluation of deep learning explanation methods,” in *Pattern Recognition and Artificial Intelligence: Third International Conference, ICPRAI 2022*, ser. Lecture Notes in Computer Science. Springer, 2022, pp. 84–95. [Online]. Available: https://doi.org/10.1007/978-3-031-09037-0_8.

COMPLIANCE WITH ETHICAL STANDARDS

This work analyses publicly released COCO and PASCAL VOC images and annotations. No participants were recruited and no new human or animal experiments were conducted.